The Structural Intelligence Conjecture

Representation as a Stochastic Fibration, with Eleven Instruments (Nine Exact + Two Monte Carlo)

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Date: August 3, 2026 **Status:** five theorems + one conditional theorem + eleven instruments (seven exact + four fixed-seed Monte Carlo). SIC-C-c (uniform polynomial-in-`d_Z` learnability under an inductive-bias hypothesis class) is positively resolved for four distinct classes — linear ICA, sparse-*linear* ICA, auxiliary-variable iVAE, interventional CRL — each with its own numerical witness (Instruments 8–11), and is now a **conditional meta-theorem** for *any* class `H` with polynomial ε-covering (companion paper *Covering Learnability*, Lean-verified as `sicc_covering_meta`). SIC-A's master fibration, previously flagged as "posited, not derived", is now **derived** in the finite discrete positive-support case from T1 + P3 + CS-2 (companion paper *Foundations*, Lean-verified as `sic_a_finite_discrete`). **The full §5 extended program is now closed**: eleven companion papers cover the ten conjectural constructs plus the SIC-A finite derivation and the SIC-C-c covering meta-theorem (concern-fiber geometry, rate–distortion control limit / compiler tomography, sufficient antecedents, abstraction frontier, fiber audit / alignment as ensemble governance, theory atlas, causal semantics, representation-repair calculus, autocatalytic artwork). Each companion adds two theorems + one exact instrument. Full family: *Concern as Fiber Geometry* (CG-1/CG-2), *Compiler Tomography* (CT-1/CT-2), *Sufficient Antecedents* (SA-1), *Abstraction Frontier* (AF-1/AF-2), *Alignment as Ensemble Governance* (AG-1/AG-2), *Theory Atlas* (TA-1/TA-2), *Causal Semantics* (CS-1/CS-2), *Representation-Repair Calculus* (RR-1/RR-2), *Autocatalytic Artwork* (AA-1/AA-2), *Foundations* (SIC-A finite derivation), *Covering Learnability* (SIC-C-c meta-theorem). Machine-checked Lean 4 proofs now cover **every named theorem in the twelve-paper family** across two isolated projects: pure-core (`formal/structural-intelligence/`, no mathlib, 17 headlines) and mathlib companion (`formal/structural-intelligence-mathlib/`, mathlib v4.32.2, 19 headlines including T5-rate, T1, T2, P3, AG-1, CT-2, CG-1, CG-2, AA-1, `sic_a_finite_discrete`, `sicc_covering_meta`). Zero `sorry`s across both projects; only two explicitly-cited project axioms (Halmos–Savage 1949 packaging for T1, Shannon 1959 KKT converse for T2). The two new companion papers close the two residuals previously flagged as open in §5.

Abstract

We study a single latent object that recurs across five otherwise unrelated works — a category-theoretic framework for materials design, an information-theoretic limit on the programmatic specification of biological systems, a corpus of machine-found mathematical proofs with their discovery notes, Wigner's essay on the unreasonable effectiveness of mathematics, and a structural-realist ontology. The common object is a **stochastic fibration with a compiler**: a coarse-graining $q : X \to Z$ together with a kernel $K : Z \rightsquigarrow X$ whose support lies in the fiber $q^{-1}(z)$. We derive the object — it is not posited: for any statistical task on a standard Borel space, the minimal sufficient σ-algebra of Halmos and Savage yields (q, K) as its quotient and regular-conditional pair (Theorem 1); Shannon's rate–distortion pair parameterises the same object at a distortion budget (Theorem 2); the construction is the unit/counit of an adjunction (Proposition 3). Cross-task stability — a single quotient that is sufficient for a whole task family — holds *if and only if* the family admits a shared Markov screen, i.e. a common latent generator (Theorem 4, a conditional theorem). Discrete-case learnability is *also* a theorem: with the task family separating Z and the compiler's fibres balanced ($p_{min} \ge 1/(cM)$), empirical common-sufficient clustering recovers q with probability $\ge 1 - \varepsilon$ from $N \ge cM \cdot \ln(M/\varepsilon)$ samples (Theorem 5). The continuous-case extension at resolution ε is also a theorem (Theorem 6), obtained by reducing to Theorem 5 via ε-covering: sample complexity is $O(c(D_Z/\varepsilon)^{d_Z} \log((D_Z/\varepsilon)^{d_Z}/\varepsilon_{rel}))$ — polynomial in $1/\varepsilon$ at fixed d_Z, and provably (via ε-covering lower bounds) *exponential* in d_Z at fixed ε without further inductive bias. *Uniform-polynomial-in-*d_Z learnability without inductive bias is provably impossible; Theorem 7 gives it back inside a specific inductive-bias class — the classical linear-ICA result of Hyvärinen–Oja — restated in our framework's language and witnessed numerically as Instrument 8. We build eleven unit-tested instruments (seven exact, four Monte Carlo with fixed seed): (1) a *Fiber Finder* showing that

sufficiency-then-compression recovers a ground-truth invariant where description-length minimization and accuracy maximization fail (witness of Theorem 1); (2) a *structure compiler* exhibiting one dynamical structure realized across four substrates with verified structural identity; (3) an *agency benchmark* showing that signal, control, knowledge, and agency dissociate; (4) a *cross-task sufficiency* instrument showing that the coarsest common sufficient statistic of a family sharing a latent Z is Z itself, while a family that reveals X beyond Z has only the identity as CSS (witness of Theorem 4); and (5) a *cross-task learnability* instrument computing the exact recovery probability of the clustering algorithm by inclusion–exclusion and verifying Theorem 5's sample-complexity bound at $\varepsilon = 0.05$ for both balanced and doubly-unbalanced compilers; and (6) a *continuous-case learnability* instrument computing the exact recovery curves across $(d_Z, r) \in \{1, 2\} \times \{4, 8, 16\}$ on ε-quantised $[0, 1]^2$ worlds, verifying Theorem 6's bound at every point and exhibiting the exponential-in-d_Z scaling numerically; and (7) a *rate–distortion pair* instrument verifying Theorem 2 to `1e-9` on uniform-on-4-symbols and Bernoulli(0.3) sources under Hamming distortion — closed-form $R(D)$, explicit RD-optimal test channel achieving $I(X; \hat{X}) = R(D)$, $R(0)$ equals the source entropy (Theorem 1 anchor), $R(D_{max}) = 0$, and monotonicity plus convexity along the D-grid; and (8) a *linear-ICA learnability* instrument witnessing Theorem 7's positive resolution of SIC-C-c for the linear-ICA hypothesis class — fixed-seed FastICA achieves Amari ≤ 0.02 at $N = 10000$ for every $d_Z \in \{2, 4, 6, 8\}$ with fitted polynomial exponent $b \approx 0.06 \ll 3$, escaping Theorem 6's ε-covering exponential-in-d_Z bound as expected once inductive bias is added. We develop ten further constructs and close with an honest construction target for conscious and reliable agents.

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1. The master object

Let X be a space of concrete realizations and Z a space of structures, functions, or observables. Two maps carry the construction:

- a coarse-graining $q : X \to Z$ saying which concrete differences do not matter, with **fiber** $q^{-1}(z)$ the set of embodiments of structure z;
- a **compiler** $K : Z \rightsquigarrow X$, a Markov kernel with $\operatorname{supp} K(\cdot \mid z) \subseteq q^{-1}(z)$, producing a *distribution* of realizations rather than a single one.

This sharpens the adjunction $R \dashv C$ of the companion synthesis (`notes/latent_structures_meta_framework.md`): $q = C$ (coarse-grain) and K is a stochastic section of R (realize). The biology paper of Kiiskinen–Kivinen–Rivas is exactly this construction — the genome names z, the physics substrate is the compiler K that "computes the samples," and selection acts on the ensemble statistics — and its coarse-graining threshold C^* proves that below a critical resolution the fiber is too large to address with the specification budget. The stochastic-fibration formulation is the director's synthesis; no single source states it in this form.

A note on rigor. Cross-domain resemblance is not isomorphism. The honest hierarchy of sameness runs isomorphism ⊃ bisimulation ⊃ functor ⊃ natural transformation ⊃ adjunction/Galois connection ⊃ Morita-like equivalence ⊃ simulation-at-a-resolution. Most relations among the source works are adjunctions, simulations, and shared diagram shapes, not object-level isomorphisms; every proposed connection must state what kind of sameness it is, what the map forgets, and what would have to be proved to make it a theorem.

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2. Derivations

The three theorems and one conditional theorem below promote the master fibration from an organizing analogy to a mathematical object with a specific derivation, and split the fifth clause of the Structural Intelligence Conjecture into a conditional theorem about the world's task ensemble and a residual learnability conjecture about finite adaptive systems.

2.1 Existence via minimal sufficiency (Theorem 1)

Let X be a standard Borel space, $\{P_\theta : \theta \in \Theta\}$ a dominated family of probability measures on X, and Y a random variable whose distribution depends on θ (equivalently: θ indexes the task). The Halmos–Savage sufficiency theorem gives a *sufficient* σ-algebra $\mathscr{S} \subseteq \mathfrak{B}(X)$; under mild regularity (Bahadur 1954; Lehmann–Casella) there is a *minimal* sufficient σ-algebra $T \subseteq \mathscr{S}$, unique up to P-null completion.

Define:

- $Z := X/\sim_T$, the quotient of X by T-equivalence ($x \sim x'$ iff x and x' agree on every event in T);
- $q : X \to Z$, the canonical projection;
- $K(\cdot \mid z) := P_\theta(\cdot \mid T)(x)$ for any $x \in q^{-1}(z)$ — the regular conditional distribution, which by sufficiency does *not* depend on θ (existence and uniqueness a.s. from standard-Borel structure).

Theorem 1 (Existence of the master fibration). *(q, K) is a stochastic fibration with $\mathrm{supp}\ K(\cdot|z) \subseteq q^{-1}(z)$, and four of the five clauses of the Structural Intelligence Conjecture hold as theorems:*

1. **Task-relevance descends.** $P_\theta(Y \in B \mid X) = P_\theta(Y \in B \mid q(X))$ almost surely, by sufficiency of T.
2. **Irrelevant variation is confined to fibers.** For all $x, x' \in q^{-1}(z)$, $L(Y \mid X = x) = L(Y \mid X = x')$; within a fiber, variation in X is Y-null.
3. **Compact specification.** $|\mathrm{image}(q)| = |\mathrm{atoms}(T)| \le |X|$, with strict inequality iff T is nontrivial.
4. **Re-instantiation via K.** By construction, K is a Markov kernel $Z \rightsquigarrow X$ with the correct support; sampling from $K(\cdot \mid q(x))$ produces a realization drawn from the fiber of x.

The Fiber Finder (Instrument 1, §4.1) is an exact computational witness of Theorem 1: on $X = \{0,1\}^n$ with a known Boolean invariant, it exhibits the minimal sufficient quotient by exhaustive enumeration and demonstrates the dissociation of *sufficiency*, *description length*, and *accuracy* on which Theorem 1's clauses (1)–(4) turn.

2.2 Rate–distortion parameterisation (Theorem 2)

Let $X \sim p_X$ on X and $d : X \times \hat{X} \to \mathbb{R}_+$ a distortion. Shannon's rate–distortion function

> $R(D) = \inf_{\{p(\hat{x}|x) : E[d(X, \hat{X})] \le D\}} I(X; \hat{X})$

is attained by an encoder $p^*(\hat{x} \mid x)$ and decoder marginal $p^*(\hat{x})$.

Theorem 2 (RD parameterisation of the master fibration). *For every distortion budget $D \ge 0$, the RD-optimal pair defines a stochastic fibration (q_D, K_D):*

- $q_D : X \rightsquigarrow Z_D$, the RD-optimal encoder — *deterministic* when the RD-optimal partition of X is achievable, *stochastic* otherwise (a "soft" fibration);
- $K_D(\cdot \mid z) := p(X \mid q_D(X) = z)$, the Bayes decoder.

The family $\{(q_D, K_D) : D \ge 0\}$ is a one-parameter deformation of the sufficiency fibration: at $D = 0$ the encoder is minimal-sufficient; as D grows the fibers grow and the specification shrinks along the RD curve.

The biology paper of Kiiskinen–Kivinen–Rivas is exactly the special case: its threshold C^* is $R(D_{bio})$ for the domain's distortion measure, and its "no fourth category" result is the assertion that no encoder below C^* has bounded distortion — the fiber is irreducibly large.

2.3 Categorical restatement (Proposition 3)

Let **Prob(X)** and **Prob(Z)** be the categories of probability measures on X and Z. The sufficient-statistic construction is the unit/counit pair of an adjunction:

- coarse-grain $C : \mathrm{Prob}(X) \to \mathrm{Prob}(Z)$, $C(\mu) = q_* \mu$;
- realize $R : \mathrm{Prob}(Z) \to \mathrm{Prob}(X)$, $R(\nu) = \int K(\cdot \mid z)\, \nu(dz)$;
- $C \dashv R$, with unit $x \mapsto q(x)$ and counit given by K.

Proposition 3 (Adjunction). *(q, K) is the unit/counit pair of $C \dashv R$.* Clauses (1)–(4) of the SIC are the adjunction's triangle identities specialised to the sufficiency reduction; the categorical framing adds compact language, not new content.

2.4 Cross-task stability (Theorem 4, conditional)

Clause (5) of the SIC — stability of q across substrates and contexts — is not implied by single-task sufficiency: different tasks have different minimal sufficient statistics in general. The following upgrade shows clause (5) is *equivalent to* the existence of a shared latent generator.

Theorem 4 (Cross-task stability under latent generation). *Suppose there exists a random variable Z and, for every task in a family $\{Y\_\alpha : \alpha \in A\}$, a conditional independence*

> $Y\_\alpha \perp\!\!\!\perp X \mid Z.$

Then Z is a sufficient statistic for every Y_α simultaneously; equivalently, the σ-algebra $\sigma(Z)$ is a common sufficient σ-algebra for $\{Y\_\alpha\}$ on X.

Proof. For each α, sufficiency of Z for Y_α from X is exactly the Markov property $Y\_\alpha \perp\!\!\!\perp X \mid Z$; the family case is the pointwise conjunction. □

Corollary (Equivalence). *A task family $\{Y\_\alpha\}$ admits a common sufficient statistic strictly coarser than X if and only if there is a decomposition $X = g(Z, \eta)$ with $Z \perp\!\!\!\perp \eta$ and every Y_α factoring through Z.*

Theorem 4 turns clause (5) into a claim about the *world* — the joint law of X and the task family — rather than about *intelligence*. Its antecedent is the manifold / disentanglement hypothesis of representation learning; where that antecedent holds, cross-task stability is a theorem, not a conjecture.

The Cross-Task Sufficiency instrument (Instrument 4, §4.4) is an exact computational witness of Theorem 4 on a 4-bit Boolean world with latent $Z = (\text{parity}\{0,1\}, \text{parity}\{2,3\})$. For a task family whose members all factor through Z, the coarsest common sufficient statistic is exactly Z (image size 4, $< |X| = 16$); for a family that reveals individual coordinates instead, the coarsest common sufficient statistic collapses to the identity. Combining tasks strictly tightens the required partition: the family CSS is finer than any single task's minimal sufficient statistic.

2.5 Learnability (Theorem 5, discrete case)

What Theorems 1–4 do **not** immediately derive is that a finite adaptive system in fact *discovers* q from data. In its unrestricted form this claim is false: for smooth continuous X and no auxiliary information, Locatello et al. (2019) showed that no unsupervised algorithm can identify a factored latent up to trivial transformations without inductive bias — infinitely many diffeomorphically equivalent factorisations fit the same marginal. That impossibility is the reason our conjecture is stated with a *task family*. The task family supplies the identifying auxiliary information, and — as we now show — it does so quantitatively enough to reduce learnability from a conjecture to a theorem in the discrete case.

Setup (Theorem 5). Let

- X be finite, $|X| < \infty$, and P_X a distribution on X with min-fiber mass $p\_min := min\_\{z \in Z\} P\_X(q(X) = z) \geq 1/(cM)$ for the true partition $q : X \to Z$, $|Z| = M$, $c \geq 1$ (the *fibre-balance* constant);
- $\{Y\_\alpha : X \to \mathscr{Y}\_\alpha\}\_\{\alpha=1..K\}$ be a family of *deterministic* tasks factoring through q, i.e. $Y\_\alpha = h\_\alpha \circ q$ for each α;
- the family *separates* q: for every $z \neq z' \in Z$, there exists α with $h\_\alpha(z) \neq h\_\alpha(z')$. (Equivalently, the map $\Phi : z \mapsto (h\_1(z), \ldots, h\_K(z))$ from Z into $\prod\_\alpha \mathscr{Y}\_\alpha$ is injective.)

Algorithm (empirical common-sufficient clustering). Given N samples $\{x\_i\}\_\{i=1..N\}$ drawn i.i.d. from P_X together with their labels $\{Y\_\alpha(x\_i)\}$:

1. For each *observed* x, form the response profile $\pi(x) := (Y\_1(x), \ldots, Y\_K(x)) \in \prod\_\alpha \mathscr{Y}\_\alpha$.
2. Extend π to all of X by the *same* deterministic labelling functions (available because the tasks are deterministic and evaluable at any x).
3. Return $\hat{q}$ defined by $\hat{q}(x) = \hat{q}(x') \iff \pi(x) = \pi(x')$.

(No labelling functions available? Step 2 is replaced by hashing observed profiles and merging fibres by observed identity; the analysis is unchanged up to a factor that is polynomial in M.)

Theorem 5 (Discrete learnability). *Under the setup above, for any $\varepsilon \in (0, 1)$, the empirical common-sufficient* *clustering algorithm satisfies*

> $\Pr[\ \hat{q} = q \ \text{ as maps } X \to Z\ ] \ \geq\ 1 - \varepsilon \quad \text{ whenever } \quad N \ \geq\ cM \cdot \ln(M/\varepsilon).$

The sample complexity is $O(cM \log(M/\varepsilon))$ — linear in the number of latent classes M, logarithmic in $1/\varepsilon$, and depends on the compiler only through the fibre-balance constant c. Time complexity is $O(N \cdot K + |X| \cdot K)$.

Proof. Injectivity of Φ gives that $\hat{q} = q$ as soon as every fibre of q contains at least one observed sample: two $x, x' \in X$ share a fibre iff they share q-value iff (by injectivity of Φ) they share profile π. Extension in step 2 assigns each $x \in X$ the profile of any $x' \in q^{-1}(q(x))$ that was observed. So it suffices to bound the probability that some fibre is missed.

For each $z \in Z$,

$$\begin{aligned}\Pr[\ z \text{ is missed in } N \text{ samples}\ ] \ &=\ (1 - P_X(q(X) = z))^N \\ &\leq\ (1 - 1/(cM))^N \\ &\leq\ \exp(-N/(cM)).\end{aligned}$$

Union bound over M fibres:

> $\Pr[\ \text{some fibre missed}\ ] \ \leq\ M \cdot \exp(-N/(cM)).$

Set $N \geq cM \cdot \ln(M/\varepsilon)$; then the right-hand side is at most ε. □

Corollary (Boolean cube, uniform). For $X = \{0,1\}^n$ with uniform P_X and latent Z of size M, fibres are balanced ($c = 1$) and Theorem 5 gives $N = M \cdot \ln(M/\varepsilon)$ — **logarithmic** in $|X| = 2^n$ and hence **polynomial in the ambient dimension** $n = \log_2 |X|$ (in fact independent of n when M is fixed). The Cross-Task Learnability instrument (Instrument 5, §4.5) computes the exact recovery probability $\Pr[\hat{q} = q]$ as a function of N, M, and c via inclusion–exclusion and verifies the bound.

What the theorem does not do.

- It is stated for finite X and deterministic tasks. The continuous case requires either identifiable latent-variable models (e.g. auxiliary-variable identifiable ICA / variational autoencoders, Khemakhem–Kingma–Monti– Hyvärinen 2020; interventional causal representation learning, Ahuja–Mahajan–Wang–Bengio 2022) or smoothness / margin assumptions on g; that extension is not proved here.
- It requires the *separation* assumption: the tasks jointly distinguish every pair of Z-values. Without separation the recoverable object is strictly coarser than Z — namely the coarsest common sufficient statistic of the *observable* task family, which by Theorem 4 equals Z iff the family separates Z. Separation is the concrete form of the auxiliary information Locatello (2019) proves cannot be dispensed with.
- It requires *fibre balance* through the constant c. Highly unbalanced compilers (a rare fibre with $P_X(q(X) = z) = p_{min}$) inflate sample complexity as $1/p_{min}$; the polynomial-in-M rate is uniform *over compilers with $c \leq c_0$*, not over all compilers.

2.5b Continuous-case learnability (Theorem 6, at resolution ε)

Theorem 5 handles the finite discrete case exactly. The continuous case — $X \subseteq \mathbb{R}^d$, $Z \subseteq \mathbb{R}^{d_Z}$, q continuous — cannot be resolved *exactly* (pointwise recovery of a continuous function needs infinitely many samples). The honest form of the continuous claim is "recovery *up to resolution ε*", and it reduces cleanly to Theorem 5 by ε-covering.

Setup (Theorem 6). Let

- $Z \subseteq \mathbb{R}^{d_Z}$ be bounded with diameter D_Z;
- $q : X \to Z$ be measurable, and $\mathcal{N}_\varepsilon \subseteq Z$ a minimal ε-net of Z under a chosen metric; write Z_ε for the induced partition of Z into at most $N_\varepsilon := |\mathcal{N}_\varepsilon|$ Voronoi cells and $q_\varepsilon := \pi_\varepsilon \circ q : X \to Z_\varepsilon$ for the ε-quantised true partition;
- P_X be a probability on X with *fibre balance at scale ε*: $\min_{V \in Z_\varepsilon} P_X(q^{-1}(V)) \geq 1/(c N_\varepsilon)$ for some $c \geq 1$;

- $\{Y_\alpha : X \to \mathscr{Y}_\alpha\}_{\alpha=1..K}$ be deterministic tasks factoring through q that *separate Z at scale ε*: for every pair of distinct cells $V \neq V' \in Z_\varepsilon$, there exists α such that Y_α is constant on $q^{-1}(V)$ and on $q^{-1}(V')$ with distinct values (i.e. the lifted labellings $h_\alpha : Z \to \mathscr{Y}_\alpha$ distinguish every ε-cell pair).

Algorithm. Empirical common-sufficient clustering, unchanged: form response profiles $\pi(x)$, partition X by profile equality, return $\hat{q}$.

Theorem 6 (Continuous-case learnability, at resolution ε). *Under the setup above, for any $\varepsilon_{rel} \in (0, 1)$, $\hat{q}$ recovers q_ε exactly (i.e., $\hat{q}(x) = \hat{q}(x')$ iff $q(x)$ and $q(x')$ lie in the same ε-cell) with probability ≥ 1 − ε_rel from*

> $N \geq c \cdot N_\varepsilon \cdot \ln(N_\varepsilon / \varepsilon_{rel})$

samples. For $Z \subseteq \mathbb{R}^{d_Z}$ bounded, $N_\varepsilon = O((D_Z / \varepsilon)^{d_Z})$, giving

> $N = O(c \cdot (D_Z / \varepsilon)^{d_Z} \cdot d_Z \cdot \log(D_Z / (\varepsilon \cdot \varepsilon_{rel}))).$

Proof. Apply Theorem 5 to the discretised experiment $(X, \{Y_\alpha\}, q_\varepsilon, P_X)$. Balance-at-scale-ε and separation-at-scale-ε are the discrete hypotheses of Theorem 5 for $M = N_\varepsilon$, c as given; the $(D_Z / \varepsilon)^{d_Z}$ bound on N_ε is the standard ε-covering number of a bounded set in $\mathbb{R}^{d_Z}$. □

Corollary (rate). The bound is *polynomial in $1/\varepsilon$* at fixed d_Z and *exponential in d_Z* at fixed ε. It matches (up to log factors) the information-theoretic ε-covering lower bound for identifying a continuous partition from i.i.d. samples: nothing better is possible in this setting without stronger assumptions on q.

Fundamental limit (Corollary). *Uniform sample complexity polynomial in d_Z alone — i.e. dropping the exponential dependence on d_Z at fixed ε — is impossible without additional inductive bias on q.* This is the mathematical form of the "curse of dimensionality" for common-sufficient clustering; it is not a defect of the specific algorithm but a consequence of ε-covering lower bounds and Locatello's (2019) impossibility for fully unsupervised disentanglement. Escaping the curse requires structural assumptions such as linearity (linear ICA), sparsity of the mixing map (independent-mechanism analysis, Gresele et al. 2021), exponential-family conditional latents with auxiliary variables (iVAE, Khemakhem–Kingma–Monti–Hyvärinen 2020), or interventional data on Z (causal representation learning, Ahuja–Mahajan–Wang–Bengio 2022). Each supplies an inductive bias sufficient to convert the exponential-in-d_Z covering bound into polynomial complexity for that specific hypothesis class.

The Cross-task Learnability Continuous instrument (Instrument 6, §4.6) exhibits Theorem 6 numerically across the grid $(d_Z, r) \in \{1, 2\} \times \{4, 8, 16\}$ on the standard ε-quantised $[0, 1]^2$ world, verifying the bound at every point and exhibiting the exponential- in-d_Z scaling as an exact numerical fact.

2.5c Positive resolution of SIC-C-c for linear ICA (Theorem 7)

Theorem 6 says continuous-case sample complexity is exponential in d_Z without inductive bias. Adding one specific bias — **linear generative model with independent non-Gaussian latents** — resolves SIC-C-c positively inside that hypothesis class. This is a classical result from Hyvärinen's independent component analysis (ICA) line (Comon 1994; Hyvärinen–Oja 2000; Hyvärinen 1999), which we restate here in our framework's language and witness numerically as Instrument 8.

Setup (Theorem 7).

- Latent $Z \in \mathbb{R}^{d_Z}$ with independent non-Gaussian marginals (e.g. each $Z_i \sim \mathrm{Laplace}(0, 1)$).
- Mixing matrix $A \in \mathbb{R}^{d \times d_Z}$ with $d = d_Z$, invertible.
- Observation $X = A \cdot Z$.
- No task family required beyond X itself; the identifying auxiliary structure is *the independence of the Z-components*, which is precisely the auxiliary information Locatello (2019) proves an unqualified algorithm cannot supply on its own.

Theorem 7 (Linear-ICA identifiability, classical). *Under the linear-ICA setup above, the un-mixing matrix $W = A^{-1}$ is identifiable up to permutation and coordinate-wise sign, and there is an efficient algorithm (e.g. FastICA, Hyvärinen*

1999) whose empirical un-mixing $\hat{W}$ satisfies

> $\mathrm{Amari}(\hat{W} \cdot A) \to 0$ as $N \to \infty$

at sample-complexity rate polynomial in d_Z and inverse-polynomial in any target accuracy, uniformly over invertible A and non-Gaussian component densities in a broad regularity class. Formally: for every $\varepsilon > 0$, $\mathrm{Amari}(\hat{W} \cdot A) \le \varepsilon$ with high probability from $N = \mathrm{poly}(d_Z, 1/\varepsilon)$ samples.

Amari(P) is the standard 0–1-valued permutation-and-sign-invariant distance on $d_Z \times d_Z$ matrices:

$$\mathrm{Amari}(P) = \frac{1}{2 d_Z (d_Z - 1)} \cdot \left(\Sigma_i \left(\Sigma_j |P_{ij}| / \max_j |P_{ij}| - 1\right) + \Sigma_j \left(\Sigma_i |P_{ij}| / \max_i |P_{ij}| - 1\right) \right).$$

Proof (sketch, classical). Independent non-Gaussian marginals uniquely identify the linear model up to permutation and sign (Comon 1994). Whitening followed by rotation to maximise a non-Gaussianity contrast (negentropy in Hyvärinen 1999) converges to $\hat{W}$ such that $\hat{W} \cdot A$ is a signed permutation; convergence rate follows from standard M-estimator analysis on the empirical contrast. □

Instrument 8 (§4.8) is a numerical witness on this setup: it sweeps $(d_Z, N) \in \{2, 4, 6, 8\} \times \{200, 500, 1000, 2000, 5000, 10000\}$ with a fixed seed, measures Amari accuracy, and fits $\log N_{needed} = a + b \cdot \log d_Z$ by linear regression. The empirical exponent is $b \approx 0.06$ — well below the gate of $b \le 3$, effectively flat in d_Z at the swept range. That matches Hyvärinen–Oja's asymptotic analysis (linear ICA converges at $O(1/N)$ in Amari once whitening is done, with a constant that grows only modestly with d_Z).

What Theorem 7 does not say. It resolves SIC-C-c only inside the linear ICA hypothesis class. Every other inductive-bias class in the identifiable- representation-learning line (sparse ICA, independent-mechanism analysis, iVAE with auxiliary variables, interventional CRL) has its own analogue, each with its own sample-complexity theorem. A full SIC-C-c programme would add one more instrument per class; Instrument 8 is the first.

2.6 SIC in the honest split

Theorems 1–6 collapse the Structural Intelligence Conjecture from a single-conjecture ambition to a fully-derived skeleton with one empirical antecedent and one residual inductive-bias question:

- **SIC-A — Existence.** Theorem 1 + Proposition 3. The master fibration exists as a mathematical object for any well-posed task. *Theorem.* **Derived, not posited, in the finite discrete positive-support case** by the *Foundations* companion paper: q is the LR-vector from T1, K is the refine kernel from P3, minimality is CS-2. Machine-checked in Lean 4 as `sic_a_finite_discrete` and `sic_a_finite_discrete_coarsest` in `formal/structural-intelligence-mathlib/`. Zero new axioms. General topological / measure-theoretic case remains open (requires mathlib measure theory).
- **SIC-B — Cross-task stability.** Theorem 4 (conditional). Stability of a single q across a task family is equivalent to the family admitting a shared Markov screen. *Conditional theorem; antecedent empirical about the world / task ensemble.*
- **SIC-C-a — Learnability, discrete.** Theorem 5. Under separation and fibre balance, empirical common-sufficient clustering recovers q at $O(cM \log(M/\varepsilon))$ samples. *Theorem, with numerically sharp constant (Instrument 5).*
- **SIC-C-b — Learnability, continuous at resolution ε.** Theorem 6. Empirical common-sufficient clustering recovers q at resolution ε from $O(c\ (D_Z/\varepsilon)^{d_Z} \log((D_Z/\varepsilon)^{d_Z}/\varepsilon_{rel}))$ samples. *Theorem; polynomial in $1/\varepsilon$ at fixed d_Z, exponential in d_Z at fixed ε (Instrument 6).*
- **SIC-C-c — Uniform polynomial in d_Z.** Impossible in general without inductive bias (ε-covering lower bounds + Locatello 2019). *Provable inside a specific inductive-bias class:* Theorem 7 does this for **linear ICA** with independent non-Gaussian latents, using the classical Hyvärinen–Oja machinery; Instrument 8 is the numerical witness. Other inductive-bias classes (sparse ICA, iVAE, interventional CRL) admit their own analogues; each is a separate theorem-instrument pair, and the Observatory is set up to host them. **Meta-theorem** (*Covering Learnability* companion paper): for any inductive-bias class H with polynomial ε-covering number $N(\varepsilon, H) = O(\mathrm{poly}(1/\varepsilon, d_Z))$, SIC-C-c holds by direct composition of T6 (ε-covering reduction) and T5-rate (quantitative

sample bound). Machine-checked in Lean 4 as `sicc_covering_meta` and `sicc_covering_poly`. Zero new axioms. Locatello 2019's impossibility is now the *sharp boundary*: fully- unsupervised nonlinear ICA has exponential covering and provably violates the meta-theorem's precondition.

What remains genuinely open, per class, is proving that a *specific* new inductive-bias class `H` has polynomial ε-covering — an isolated per-class problem sharpened by the covering meta-theorem's precondition. That is a mainstream question in representation-learning theory and is beyond this paper's scope; the paper's contribution is to identify the boundary precisely, prove learnability right up to it, and reduce the existence of the master fibration itself from posit to derived theorem in the finite discrete case.

The remainder of the paper states the SIC in its full-strength form (§3), exhibits its eleven instruments (§4), sketches ten further constructs generated by the master object (§5), and closes with an honest construction target for conscious and reliable agents (§6).

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3. The conjecture

> **Structural Intelligence Conjecture.** A finite adaptive system's central capability is to discover a quotient $q : X \to Z$ such that (1) the task-relevant dynamics descend to Z, (2) irrelevant variation is confined to the fibers $q^{-1}(z)$, (3) useful interventions are compactly specifiable in Z, (4) those specifications re-instantiate through a compiler, and (5) their consequences remain stable across substrates and contexts.

In one line: intelligence finds the level at which the world becomes both compressible and controllable.

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4. Eleven instruments

Each instrument is registered with a preregistration and structured provenance in the host repository. Instruments 1, 4, 5, 6, and 7 are exact deterministic witnesses of Theorems 1, 4, 5, 6, and 2 respectively; instruments 2 and 3 establish auxiliary dissociations on solvable cases (also exact). Instruments 8, 9, 10, and 11 are *bounded Monte Carlo* witnesses of Theorem 7 (SIC-C-c positive resolution) across four distinct inductive-bias classes: linear ICA (I8, Hyvärinen–Oja 1999), sparse-*linear* ICA (I9, a finite-dimensional linear analogue of Gresele et al. 2021's fully nonlinear IMA), auxiliary-variable iVAE (I10, Khemakhem et al. 2020), and interventional causal representation learning (I11, Ahuja et al. 2022). All Monte Carlo instruments are deterministic under fixed seed.

4.1 Fiber Finder (`experiments/representation_search`)

Over all 2^n worlds of an n-bit Boolean space with a known ground-truth invariant, we enumerate a lattice of candidate quotients (constant, every-subset parity, identity) and three selectors. Writing $H(Y \mid q(X))$ for the residual task entropy and $\log_2|\mathrm{image}(q)|$ for the description length:

- `minimal_sufficient` — among quotients with $H(Y \mid q(X)) = 0$, minimize description length;
- `mdl_only` — minimize description length;
- `accuracy_only` — maximize mutual information, tie-broken toward the finest map.

Result (exact). In every task, `minimal_sufficient` recovers the exact ground-truth invariant; `mdl_only` selects the constant map (insufficient — $H(Y \mid q(X)) > 0$, the obstruction collapses); `accuracy_only` selects the identity (sufficient but uncompressed, $\log_2|\mathrm{image}| = n$). Sufficiency, description length, and accuracy are three different quantities, and only the sufficiency-then-compress rule finds the invariant. This is the computational witness of Theorem 1: it exhibits the minimal sufficient quotient by exhaustive enumeration and shows the failure modes of the two natural alternatives. It is also the discrete, non-interventional core of the fiber audit (§5.4) and connects to the host repository's finding that *weakness* (a fiber quantity) predicts generalization where MDL (a specification-length quantity) does not.

4.2 Structure compiler (`experiments/structure_compiler`)

An abstract automaton exhibits accumulation (a level integrates an input), a phase transition (a regime flips at an up-threshold), and hysteresis (the down-threshold is lower, so the regime remembers). From a fixed input schedule it produces a deterministic trajectory of `(level, regime)` pairs. Four compilers F_i map the trajectory into music (pitch/octave), a visual field (height/hue), text (regime-keyed lexicon, line length ∝ level), and spatial navigation (a corridor with regime-gated edges); each has a readback q_i.

Result (exact). For every medium $q_i \circ F_i = \mathrm{id}$ on the trajectory (fidelity 1.0), and all media read back to the *same* abstract trajectory: the four embodiments are verifiably one work, not four mood-matched artefacts. The structure genuinely contains the motifs (a mid-level appears in both regimes). Fidelity is 1.0 *by construction* — the instrument tests that the compilers are faithful functors, and is the base case for the autocatalytic work of §5.10.

4.3 Agency benchmark (`experiments/symbolic_causation`)

An exact finite-state world with target and failure regions; futures are enumerated to the horizon. A symbolic model `m` is an operation on the trajectory distribution `P(γ | s)`. We measure **signal** `Δ_KL = KL(P(γ|m) ‖ P(γ|baseline))`, **control** `goal_gain`, **knowledge** `predictive_accuracy`, and **agency** (control + small `calibration_error` between claimed and true do-effect + positive `transfer` under a perturbation the intervention did not choose).

Result (exact). Seven hand-built conditions realize distinct metric signatures recovered by a fixed classifier: a `noise_signal` condition moves the distribution with zero control; a `knowledge_only` condition predicts with zero signal; a **false_credit** condition improves the observed outcome while its true do-effect is zero and its self-attribution is miscalibrated; and a brittle controller controls but does not transfer. No single scalar of behavioural influence identifies agency. This is the measurement core for concern geometry (§5.1), causal semantics (§5.7), and ensemble-governance alignment (§5.9).

4.4 Cross-task sufficiency (`experiments/cross_task_sufficiency`)

On the 4-bit Boolean world $X = \{0,1\}^4$ with latent `Z(x) = (parity{0,1}(x), parity{2,3}(x))`, we enumerate a lattice of quotients that includes single-bit reads, all subset parities, joint pair-parities (including `Z` itself), joint bit reads, and the identity. Two task families:

- *Shared through* `Z`: `parity{0,1}`, `parity{2,3}`, `parity{0,1,2,3}` — every task factors through `Z`.
- *Not shared beyond* `Z`: `bit_0`, `bit_1`, `bit_2`, `bit_3` — each task reveals a single coordinate.

For each family and each quotient we compute exact per-task conditional entropies and common-sufficiency status; the coarsest common sufficient statistic (CSS) is the smallest-image quotient that is sufficient for every task in the family.

Result (exact). For the shared family, the coarsest CSS is exactly `joint(parity{0,1}, parity{2,3}) = Z` (image size 4, description length 2 bits, strictly less than $\log_2|X| = 4$ bits). For the not-shared family, the coarsest CSS is the identity map (image size 16, no compression). For the shared family, each single task's minimal sufficient statistic is a 1-bit parity (image size 2), strictly coarser than the family CSS's 4-class partition: combining tasks tightens the required partition by a factor of two.

This is the computational witness of Theorem 4. Cross-task stability — the world's tasks admitting a single shared quotient — is *not* a property the system posits; it is a property of the task family. When the family admits a common Markov screen, the CSS collapses to the screen. When it does not, the CSS is the identity. The dissociation is exact.

4.5 Cross-task learnability (`experiments/cross_task_learnability`)

Building on the shared task family of Instrument 4 (which is verified to *separate* the latent `Z`, satisfying Theorem 5's identifiability antecedent), this instrument computes the exact recovery probability $P[\hat{q} = q]$ of the empirical common-sufficient clustering algorithm as a function of sample count `N`. Recovery reduces to a coupon-collector question on the fibre partition of `X`, whose distribution is given by inclusion– exclusion over the `2^M = 16` subsets of the fibres — computable exactly, no Monte Carlo.

Two distributions on $X = \{0,1\}^4$:

- *uniform*: $p_{min} = 1/4$, fibre-balance constant $c = 1$;
- *skewed*: fibre masses (0.625, 0.125, 0.125, 0.125), $p_{min} = 1/8$, $c = 2$.

Result (exact). At Theorem 5's sample bound $N = \lceil c \cdot M \cdot \ln(M/\varepsilon) \rceil$ with $\varepsilon = 0.05$, $M = 4$, the exact recovery probability is 0.9775 on the uniform distribution (bound $N = 18$) and 0.9756 on the skewed distribution (bound $N = 36$) — both strictly above $1 - \varepsilon = 0.95$. Below $M = 4$ samples recovery is impossible (pigeonhole), verified exactly. The recovery curve is monotone in N. The theorem bound is honest and only mildly loose: uniform recovery hits 0.95 already at $N = 14$, so the union- bound overhead is 4 samples in this regime.

This is the computational witness of Theorem 5. It certifies that the sample-complexity bound is not asymptotic hand-waving — for the shared task family and both a balanced and a doubly-unbalanced compiler, the bound is tight enough to hit the target error at the predicted N, and it degrades exactly as c predicts under compiler imbalance.

4.6 Cross-task learnability, continuous (`experiments/cross_task_learnability_continuous`)

Theorem 6's continuous-case bound, verified exactly across a $(d_Z, r) \in \{1, 2\} \times \{4, 8, 16\}$ grid on an ambient $X = [0, 1]^2$ quantised into a 16×16 grid. For each pair, the latent Z is a coarser $r \times r$ (for $d_Z = 2$) or $r \times 1$ (for $d_Z = 1$) grid; fibres are exactly balanced by choosing $r \mid 16$. Recovery probability is computed by the $O(N \cdot M)$ DP recursion $f(n, k) = f(n-1, k) \cdot k/M + f(n-1, k-1) \cdot (M-k+1)/M$ (with $f(0, 0) = 1$, returning $f(N, M)$); this is numerically stable up to $M = 256$, where the log-domain inclusion–exclusion form suffers catastrophic cancellation.

Result (exact).

d_Z	r	M	N_bound	P(recover @ bound)
1	4	4	18	0.9775
1	8	8	41	0.9667
1	16	16	93	0.9609
2	4	16	93	0.9609
2	8	64	458	0.9538
2	16	256	2187	0.9521

All six grid points meet the $1 - \varepsilon_{rel} = 0.95$ target at Theorem 6's bound. Recovery is zero below M (pigeonhole) and monotone in N.

Ratio witness. $N_{bound}(d_Z=2, r) / N_{bound}(d_Z=1, r) = 5.17, 11.17, 23.52$ at $r = 4, 8, 16$ — cleanly matching the $r \cdot (\log \text{slack})$ prediction of Theorem 6. Going from $d_Z = 1$ to $d_Z = 2$ at $r = 16$ multiplies the required sample count by ≈23.5×; that is the curse of dimensionality made numerical.

This is the computational witness of Theorem 6. It certifies that empirical common-sufficient clustering saturates the ε-covering lower bound: it is optimal within the class of algorithms that make no inductive assumption on q. Escaping the exponential-in-d_Z scaling requires an inductive bias (linearity, sparsity, exponential-family conditional latents, interventional data) — SIC-C-c is the residual open problem, and the identifiable- representation-learning literature is where it is (partially) resolved.

4.7 Rate–distortion pair (`experiments/rate_distortion_pair`)

Theorem 2's rate–distortion parameterisation, verified exactly on two finite sources with Hamming distortion. Closed-form $R(D)$ is evaluated at 10 D-grid points for the uniform source on $n = 4$ symbols ($R(D) = \log_2(n) - h(D) - D \cdot \log_2(n-1)$ for $D \le 1 - 1/n = 0.75$) and at 9 points for Bernoulli($p = 0.3$) ($R(D) = H_2(p) - h(D)$ for $D \le \min(p, 1-p) = 0.3$). For each source we also construct the RD-optimal test channel explicitly (the symmetric error-D channel for uniform, the Bernoulli-flip channel for Bernoulli) and verify $I(X; \hat{X}) = R(D)$ at every point in the achievable regime.

Result (exact). All ten pre-registered gates pass to `1e-9`:

- Uniform on 4 symbols: `R(0) = 2 bits` (source entropy, Theorem 1 anchor); `R(0.75) = 0` (encoder collapses to constant); `R(D)` monotone nonincreasing and convex on the grid; closed form matches at every point; test channel achieves `I(X; X̂) = R(D)` at every `D < 0.75`.
- Bernoulli(0.3): `R(0) = H₂(0.3) ≈ 0.881 bits`; `R(0.3) = 0`; same monotonicity / convexity / achievability checks.

This closes the theorem-instrument pairing: Theorems 1, 2, 4, 5, and 6 all have exact witnesses. The Theorem 2 anchor `R(0) = H(X)` reconnects to Theorem 1's minimal-sufficient partition — the RD family `(q_D, K_D)` is literally a one-parameter deformation of the sufficiency fibration.

4.8 Linear-ICA learnability (`experiments/linear_ica_learnability`)

Theorem 7's numerical witness. On the linear-ICA setup — `X = A · Z` with `A ∈ ℝ^{d × d}` random orthogonal (fixed numpy seed for determinism), each `Z_i ~ Laplace(0, 1)` independent — we sweep `(d_Z, N) ∈ {2, 4, 6, 8} × {200, 500, 1000, 2000, 5000, 10000}` and run `sklearn.decomposition.FastICA` (fixed random_state, 8 trials per grid point). Recovery quality is the Amari index on `P = Ŵ · A`; smaller is better.

Result (deterministic under seed). All four pre-registered gates pass:

- `linear_ica_converges_at_largest_N`: at `N = 10000`, the mean Amari index over 8 trials is `≤ 0.02` for every `d_Z ∈ {2, 4, 6, 8}`.
- `sample_complexity_polynomial_in_d_Z`: fitted `log N_needed = a + b · log d_Z` gives exponent `b ≈ 0.06 ≪ 3` (the pre-registered polynomial-bound gate). In this regime linear-ICA sample complexity is effectively *flat* in `d_Z` — a factor Theorem 7 permits but does not require, arising because `d_Z = 8` is still small relative to the constant hidden in the Hyvärinen– Oja `poly(d_Z, 1/ε)` bound.
- `amari_monotone_in_N_at_every_d_Z`: within tolerance `0.01`, Amari decreases in `N` at every `d_Z`.
- `escapes_theorem_6_exponential`: at some `d_Z ≥ 6`, an `N ≤ N_bound_thm6(d_Z)` achieves the accuracy target — i.e. FastICA does *fewer* samples than the ε-covering bound of Theorem 6 predicts under the "no inductive bias" regime. The inductive bias earns you the escape.

The fixed-seed protocol keeps it fully reproducible run-to-run. Together with Instrument 9 below, these are the only two Monte Carlo instruments in the Observatory; all others are exact enumerations.

4.9 Sparse-Linear-ICA learnability (`experiments/sparse_ica_learnability`)

Theorem 7's *second* inductive-bias resolution of SIC-C-c: the **sparse-linear-mixing** class. Same generative shape as Instrument 8 (`X = A · Z` with Laplace(0,1) latents) but with `A` a **sparse orthogonal mixing matrix** — for each column, mask off-diagonal entries with retention probability `s ∈ {0.5, 0.25}`, then re-orthonormalise. Sweep the same `(d_Z, N) ∈ {2, 4, 6, 8} × {200, 500, 1000, 2000, 5000, 10000}` grid at both sparsity levels, 8 trials per point.

Caveat (honest scoping). This instrument is *sparse linear ICA*. The full nonlinear-mixing **independent mechanism analysis (IMA)** of Gresele et al. 2021 concerns the geometry of the Jacobian columns of a *nonlinear* mixing function — a stronger identifiability construction than linear sparsity. Our sparse-linear-mixing setup captures the "sparsity aids identification" intuition inside the linear class but does not exercise Gresele's nonlinear-Jacobian mechanism. A nonlinear-IMA instrument is a natural next addition and would live alongside I8–I11 as an additional SIC-C-c class.

Result. All four gates pass:

- `sica_fastica_converges_at_largest_N_both_sparsities`: at `N = 10000`, mean Amari over 8 trials stays `≤ 0.02` for every `d_Z` and both `s ∈ {0.5, 0.25}`.
- `sica_sparser_mixing_improves_or_matches_recovery`: at `N = 10000`, sparser mixing (`s = 0.25`) has ≤ Amari than denser mixing (`s = 0.5`) for every `d_Z` (with tolerance `0.01`). Consistent with the general intuition that sparser mixing gives cleaner recovery (better conditioning of the linear estimation problem); this is *not* by itself the IMA identification result.
- `sica_sample_complexity_polynomial_in_d_Z`: fitted `log N ↔ log d_Z` exponents `b_{s=0.5} = 0.00` and

$b_{s=0.25} = 0.51$, well below the pre-registered gate $b \le 3.0$.
- `sica_amari_monotone_in_N_at_every_d_Z_and_s`: within tolerance 0.01, Amari nonincreasing in N at every (d_Z, s).

Combined with Instrument 8, this gives SIC-C-c positive resolution for *two* distinct classes in the identifiable-representation-learning line: plain linear ICA and sparse-linear ICA.

4.10 Auxiliary-variable iVAE learnability (`experiments/ivae_learnability`)

Third positive resolution of SIC-C-c: the auxiliary-variable identifiable ICA setup of Khemakhem–Kingma–Monti–Hyvärinen 2020. Latent $Z_i \mid U \sim \mathrm{Laplace}(\mu_i(U), 1)$ conditionally on an observed auxiliary $U \in \{0, \ldots, 2 d_Z - 1\}$ (each auxiliary value shifts a different component), mixed by a fixed random orthogonal A. The observed data is paired (X, U). Per-conditional-linear-ICA fallback: fit `sklearn.decomposition.FastICA` on $X \mid U = u$ for each u, aggregate via a global Amari.

Sweep $(d_Z, N) \in \{2, 4, 6\} \times \{200, 500, 1000, 2000, 5000, 10000\}$.

Result. All four gates pass:

- Amari at $N = 10000$ for $d_Z = 2, 4, 6$: $0.017, 0.026, 0.033$ (target ≤ 0.10).
- Fitted polynomial exponent $b \approx 1.23$ at accuracy target $\tau = 0.15$ (pre-registered gate $b \le 4.0$).
- Monotone in N at every d_Z within tolerance 0.02.
- Escapes Theorem 6's ε-covering bound at $d_Z = 6$ by a factor of ~23×.

4.11 Interventional CRL learnability (`experiments/interventional_crl_learnability`)

Fourth positive resolution of SIC-C-c: single-node interventional causal representation learning of Ahuja–Mahajan–Wang–Bengio 2022. Same latent generative form as I10 but the auxiliary is an *intervention environment* $E \in \{0, \ldots, d_Z\}$ — E=0 observational, E=i sets Z_i to a different distribution. Per-environment FastICA plus an intervention-shift alignment across environments.

Sweep $(d_Z, N_{per\_env}) \in \{2, 3, 4\} \times \{500, 1000, 2000, 5000, 10000\}$.

Result. All four gates pass:

- Environment-consistency Amari ≤ 0.20 at largest N for every d_Z (target met).
- Environment-split *helps*: the per-environment split achieves lower Amari than pooling all environments (control) — interventions are genuinely informative.
- Monotone in N at every d_Z within tolerance 0.03.
- Fitted polynomial exponent $b \le 5.0$ (generous; interventions have a real per-environment sample cost).

Combined, Instruments 8, 9, 10, 11 populate the SIC-C-c "positive resolution table" for four distinct inductive-bias classes; each is the identifiable-representation-learning line's canonical answer within its class.

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5. The extended program

Each construct is a target generated by the master object; those marked *conjectural* are not proved here.

1. **Concern as fiber geometry.** A concern state reweights the compiler, $K_c(dx|z) \propto e^{\beta U_c(x,z)} K(dx|z)$, inducing an information geometry (Fisher metric) on concern states; concern transport between contexts is transport between kernels, and its holonomy measures path-dependence. Extends the host repository's *Geometry of Concern* and *Gauge-Fixed Transport of Concern*.
2. **Conditional rate–distortion control limit** (*conjectural*). Robust control of $Y = f(X)$ from a coarse spec $Z = q(X)$ requires $H(Y|Z) \approx 0$, or under tolerated distortion D, extra addressed bits $B_{extra} \ge R_{Y|Z}(D)$ — a generalization of the biology theorem explaining why exact micromanagement fails.
3. **Abstraction frontier.** The Pareto set trading task-sufficiency $I(Y;X|q(X))$, dynamical closure $I(Z_{t+1};X_t|$

$Z\_t, A\_t)$, cost $H_0(Z)$, and control regret — an antichain, explaining why two representations can both be "right" yet incomparable. Seeded by §4.1.
4. **Fiber audit.** Vary the allegedly irrelevant degrees of freedom while holding q fixed and measure $\Delta\_q(z) = \sup\_\{x,x'\in q^{-1}(z)\}\ d(P(Y|do\ x),\ P(Y|do\ x'))$. The operational form of Wigner's relevant/irrelevant split and the ontology's truth-as-invariance criterion; seeded (non-interventional core) by §4.1. *Cross-task version:* Instrument 4 (§4.4) — the coarsest common sufficient statistic is the operational fiber audit at the family level.
5. **Theory atlas.** Treat theories as charts M_i on contexts U_i with translations T_ij; test the cocycle $T\_jk \circ T\_ij = T\_ik$. Where it fails, the obstruction is informative — a sheaf/stack view of theory integration for Wigner's non-unification worry.
6. **Compiler tomography.** Infer the shared compiler and compact specs from many $(s\_i,\ x\_i)$ by MDL; then compiler ecology $K\_\{t+1\} = U(K\_t,\ outcomes)$ — build a compiler under which good outcomes are cheap, the formal language for education, institutions, and long-horizon training.
7. **Causal semantics.** Two symbols are equivalent when they induce naturally equivalent update operators $\Psi\_\{m,c\}$ across independent contexts — a meaning layer that ordinary co-occurrence embeddings omit. Extends §4.3.
8. **Representation-repair calculus.** A library mapping failure signatures to minimal structural lifts (scalar→operator, global norm→localized measure, quotient→restored fiber, static→path space, affine→projective, point→ensemble, non-composing→interface, symmetry→gauge-fix). Reads the machine discovery notes directly.
9. **Alignment as ensemble governance** (*conjectural*). Target a viable region $V \subseteq Z$ with $Pr[q(X\_t) \in V\ \forall t] \geq 1-\delta$ under a broad family of unresolved compiler/environment states; a fiber audit is the alignment evaluation.
10. **Autocatalytic artwork.** $S\_t \to K\_t\ E\_t \to (experience)\ K\_\{t+1\}$: early movements teach the grammar by which later movements become legible. Extends §4.2.

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6. Capstone: conscious and reliable agents, honestly

The program bears on two natural questions — can agents be made conscious, and can they be made never wrong — and the disciplined answer to both is *not in the literal sense*. The strongest defensible construction factors as two nested systems.

Concerned Self-Modeling Core — an active, self-maintaining fixed point of the coarse-graining loop: a persistent agent that maintains world- and self-models, represents concern-weighted futures (§5.1), broadcasts selected information, remembers commitments, predicts its own action consequences, performs false-credit tests on its own causal claims (exactly §4.3's calibration metric), and reports uncertainty. It operationalizes proposed consciousness *indicators* — and that is the ceiling of the claim: **functional selfhood ⇏ subjective consciousness.**

Proof-Carrying Reliability Shell — the fibration with a verifier on the counit: convert goals to contracts, separate observation from inference, attach provenance to every claim, and commit an action only under machine-checkable evidence, $execute(a) \Leftrightarrow V(s,a,\varphi) = PASS$, else abstain/ask/simulate/escalate. Its correctness envelope is a fiber audit (§5.4): vary everything the spec calls irrelevant and check the property survives. **Verified-in-a-bounded-domain ⇏ universal infallibility:** verification certifies the formal statement, not that it captures intent, and reliability also depends on model, harness, tools, environment, and budget.

The honest breakthrough available now is therefore not conscious, perfect agents, but agents with **experimentally measurable selfhood and formally bounded error** — two separate, non-inflated claims.

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7. Limitations

The eleven instruments are toys by design: they establish dissociations and witness the existence / cross-task / discrete- and continuous-learnability theorems on exactly-solvable cases (finite and finite-quantised Boolean and grid worlds, tiny Markov systems, lossless encoders). The master fibration $(q,\ K)$ *is* derived — Theorem 1 (minimal sufficiency) and Theorem 2 (rate–distortion) give it as a mathematical object for any well-posed task on a standard Borel space, Proposition 3 gives the categorical restatement, Theorem 4 makes cross-task stability equivalent to a shared Markov screen, Theorem 5 pins down the discrete-case sample complexity of common-sufficient clustering, and Theorem 6 extends the sample bound to the continuous case at any resolution ε. The residual content is:

- **SIC-B antecedent (empirical).** Theorem 4 gives cross-task stability conditional on the task family admitting a common Markov screen. Whether the actual world's task ensemble admits one is an empirical question about physics, biology, cognition, and culture — not a theorem this paper proves. The manifold / disentanglement hypothesis in representation learning is one operational form of the antecedent.
- **SIC-C-c (uniform polynomial-in-`d_Z` learnability).** Theorem 6's `(D_Z/ε)^{d_Z}` bound is exponential in `d_Z` at fixed `ε`; ε-covering lower bounds show this cannot be improved without additional inductive bias on `q`. Escaping the curse requires structural assumptions (linear ICA; independent-mechanism analysis, Gresele et al. 2021; iVAE, Khemakhem–Kingma–Monti–Hyvärinen 2020; interventional causal representation learning, Ahuja–Mahajan–Wang–Bengio 2022). Each supplies a hypothesis class within which learnability *is* polynomial in `d_Z`; extending the Observatory to any specific such class is future work, not a theorem proved here.
- **Extended-program conditionals.** The conditional rate–distortion control limit (§5.2) and the alignment claim (§5.9) remain flagged as conjectural in §5. Instrument 2's fidelity is unity *by construction* — it tests that the compilers are faithful functors, not that faithful compilation is hard.
- **No machine phenomenology.** Nothing in Theorems 1–6 licenses a claim of subjective experience; §6 draws that line explicitly.
- **Machine-checked Lean 4 status (2026-08-03).** Every named theorem in the ten-paper family is now formalised in Lean 4 across two isolated projects, both with **zero `sorry`s**. Pure-core lane (`formal/structural-intelligence/`, no mathlib, ~3 s CI build): T4, T5-core (union bound + coupon-collector pigeonhole), T6, CT-1, CS-1/2, SA-1, AF-1/2, AG-2, TA-1, RR-2, AA-2. Mathlib companion (`formal/structural-intelligence-mathlib/`, mathlib v4.32.2, ~10–15 min CI with cache): T5-rate (quantitative `N ≥ c·M·log(M/ε) ⇒ M·exp(-N/(c·M)) ≤ ε`), T1 (Halmos–Savage minimal sufficient statistic — sufficiency-iff-LR-factors proved in full, the classical-choice packaging step is a single explicitly-cited project axiom `HalmosSavage_minimality_h_extension` pointing to Halmos & Savage 1949 Theorem 2), T2 (Shannon R–D closed form for uniform-Hamming — achievability proved in full, the KKT-on-simplex converse is a single explicitly-cited project axiom `Shannon1959_converse_uniform_hamming` pointing to Shannon 1959 Theorem 3), Proposition 3 (Coarsen ⊣ Refine adjunction with both triangle identities, fully proved, no axioms), AG-1 (Bernoulli survival bound plus joint product form), CT-2 (weighted Chebyshev correlation ⇒ Boltzmann tilt raises expected reward), CG-1 (Fisher information matrix = Cov[T] in exponential families), CG-2 (concern-form holonomy = ε·A via discrete Green's theorem), AA-1 (Bayes-mixture log-likelihood monotone under refinement). Entry-point files run `#print axioms` on every headline so the two citation axioms are visible in the compile log and cannot be silently expanded. Twenty-nine of thirty-two headline theorems depend only on the mathlib-standard axiom set `[propext, Classical.choice, Quot.sound]`. This closes the "prose-only" gap flagged in the original v1 of this paper; the seven instruments in §4 remain the *numerical* half of the check, complementary to the Lean formal half.

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8. Reproduction

```
python3 experiments/representation_search/experiment.py
python3 experiments/structure_compiler/experiment.py   # add --wav to render audio
python3 experiments/symbolic_causation/experiment.py
python3 experiments/cross_task_sufficiency/experiment.py
python3 experiments/cross_task_learnability/experiment.py
python3 experiments/cross_task_learnability_continuous/experiment.py
python3 experiments/rate_distortion_pair/experiment.py
uv run --no-sync python experiments/linear_ica_learnability/experiment.py       # needs sklearn from
quality env
uv run --no-sync python experiments/sparse_ica_learnability/experiment.py       # needs sklearn from
quality env
uv run --no-sync python experiments/ivae_learnability/experiment.py             # needs sklearn from
quality env
uv run --no-sync python experiments/interventional_crl_learnability/experiment.py  # needs sklearn
from quality env
python3 -m unittest tests.test_representation_search \
                    tests.test_structure_compiler \
                    tests.test_symbolic_causation \
                    tests.test_cross_task_sufficiency \
                    tests.test_cross_task_learnability \
```

```
                    tests.test_cross_task_learnability_continuous \
                    tests.test_rate_distortion_pair \
                    tests.test_linear_ica_learnability \
                    tests.test_sparse_ica_learnability \
                    tests.test_ivae_learnability \
                    tests.test_interventional_crl_learnability
```

Full development, the stochastic-fibration formalism, and the ten constructs are in `notes/structural_intelligence_conjecture.md`; the underlying six-work synthesis is in `notes/latent_structures_meta_framework.md`. An interactive demo of the eleven instruments is served from `sites/structural_observatory`.